SARScope

Retrospective land-cover analysis from Sentinel-1 radar

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HELD-OUT TEST OA	HELD-OUT TEST mIoU	MONTHLY MAPS
74.17%	42.39%	12 / 12 completed

Abstract

SARScope converts a geographic location and a historical start month into twelve land-cover maps, class-area summaries and predicted change tables. A U-Net with a ResNet34 encoder uses radar backscatter and annual radar statistics to distinguish nine classes. Training uses Dynamic World monthly weak labels; optical observations are not model inputs. Whole geographic regions are separated between training, validation and test sets.

The frozen seed-42 FP32 checkpoint reproduces its validation confusion matrix exactly and achieves 42.39% mean intersection-over-union (mIoU) and 74.17% overall accuracy on the held-out test set. A complete Mannheim 2021 case demonstrates retrieval, inference and reporting. Its annual agreement with ESA WorldCover is 26.30% mIoU, revealing substantial overprediction of built-up land despite 80.94% overall agreement.

Contribution and intended use

The contribution is an auditable end-to-end prototype: consistent radar preprocessing, a reproducible segmentation baseline, explicit missing-data accounting and interpretable exports. The work demonstrates technical feasibility; it does not establish state-of-the-art accuracy or validated detection of real land-use change.

Evidence boundary

This report uses saved evaluation JSON, acquisition records, monthly tables, rendered maps and the executed results notebook returned by the server. Model selection remains frozen. The interface demonstration and final assessor-access checks are separate delivery steps.

01 Data and supervision

The 2021 dataset contains 1,300 tiles from 52 regions. Each tile covers 256 × 256 pixels on a fixed 10 m grid. Splitting by whole region reduces direct geographic overlap between development and testing; it does not remove every source of spatial or label dependence.

Split	Tiles	Regions	Purpose
Training	925	37	Fit weights and normalization
Validation	200	8	Select checkpoint and compare variants
Test	175	7	Frozen final evaluation

Radar processing

The pipeline queries COPERNICUS/S1_GRD_FLOAT for IW acquisitions containing VV and VH. Earth Engine supplies calibrated imagery with upstream orbit, noise and geometric terrain processing [1]. The inherited project recipe adds angular volume-model terrain flattening with Copernicus DEM, layover/shadow masking and a 31-45° incidence-angle filter. Both orbit directions contribute to linear-power monthly means.

A local Lee-style 5 × 5 filter uses an inherited ENL × observation-count approximation before conversion to decibels. This is a processing heuristic, not a newly validated physical noise model. Invalid observations remain masked. Prepared arrays store dB inputs in float16; the final reference model uses FP32 computation. Training-only statistics normalize the features.

Weak labels and imbalance

Dynamic World is a 10 m nine-class product derived from Sentinel-2 [2]. Monthly mean class probabilities are converted to argmax labels on the radar grid. Only jointly valid radar/label pixels are scored; no extra confidence threshold is applied. Agreement with these labels is not equivalent to accuracy against field observations.

Class	Training pixel-months	Share
water	39,732,632	6.59%
trees	80,607,562	13.37%
grass	23,474,225	3.89%
flooded vegetation	2,150,173	0.36%
crops	224,277,256	37.21%
shrub and scrub	27,170,752	4.51%
built	95,575,729	15.86%
bare	72,389,706	12.01%
snow and ice	37,302,932	6.19%

A ground pixel can occur in several months. Flooded vegetation contributes only about 0.36% of scored training pixel-months.

02 Model and experimental protocol

The architecture combines a U-Net segmentation decoder [4] with a ResNet34 encoder [5], initialized with ImageNet weights during training. The contribution here is its radar-specific inputs and reproducible workflow, not a new backbone architecture.

Input group	Channels	Meaning
Current month	3	VV, VH and VV minus VH
Annual context	4	VV/VH mean and standard deviation across 12 months
Observation mask	1	Valid input support

Annual statistics make the model temporally informed, but it is not a recurrent network or temporal-attention model. Later months in the requested window influence the context. The system therefore supports retrospective analysis, not real-time forecasting.

Setting	Reference configuration
Optimizer / loss	AdamW / unweighted cross entropy; ignore invalid labels
Learning rate / weight decay	0.0003 / 0.0001
Schedule / batch size	Cosine decay / 8
Training budget / seed	30 epochs / 42
Selected checkpoint	Epoch 13, selected by validation mIoU
Numerics	FP32; AMP and TF32 disabled
Augmentation	No geometric augmentation; seeded batch shuffle

Selection and reproducibility

Earlier mixed-precision runs exposed nonfinite validation loss. Recovery was followed by a separate FP32 control run. The submitted reference is the retained unweighted FP32 checkpoint. Hash checks bind the checkpoint to the recorded dataset, and an exact validation-confusion replay passes before test evaluation. Existing test outputs are reused on subsequent runs; they are not used to select a new checkpoint.

Training trajectory

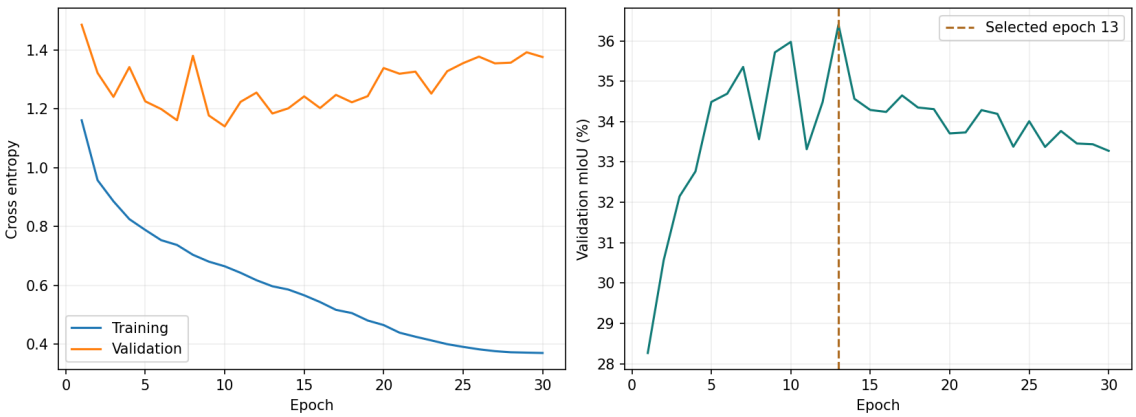


Figure 1. Preserved training history. Falling training loss alone does not establish better generalization; selection uses validation mIoU.

03 Held-out evaluation

Split	Overall accuracy	mIoU	Scored pixel-months
Validation	62.57%	36.40%	122,508,155
Test	74.17%	42.39%	112,681,019

Class	Validation IoU	Test IoU	Test F1
water	81.84%	30.55%	46.81%
trees	52.60%	62.36%	76.82%
grass	13.59%	12.84%	22.77%
flooded vegetation	0.04%	0.0047%	0.0095%
crops	52.76%	63.59%	77.74%
shrub and scrub	24.84%	18.01%	30.52%
built	69.69%	72.46%	84.03%
bare	28.23%	73.89%	84.98%
snow and ice	4.01%	47.78%	64.67%

IoU = TP / (TP + FP + FN); F1 = 2TP / (2TP + FP + FN). OA is the fraction of scored pixels classified correctly. Mean IoU averages defined class IoUs. A class absent from both prediction and reference is undefined; a false-positive-only class receives zero IoU.

Interpretation

Built, bare and crops have the strongest test IoUs. Flooded vegetation is effectively unresolved, while grass and shrub/scrub remain weak. The test score being higher than validation reflects different regions and class mixtures; it is not evidence that the model improved after selection.

Paired class-balancing experiment

Validation summary across seeds 42, 123, 2026	Result
Unweighted mean mIoU	36.446%
Weighted mean mIoU	36.706%
Paired mean difference / sample SD	+0.260 / 1.129 percentage points
Pairs with improved mIoU	1 of 3
Mean OA difference	-2.198 percentage points

These three pairs provide inconclusive descriptive evidence. The weighted variant was not promoted, and no statistical significance or reliable improvement is claimed.

04 Error structure and geography

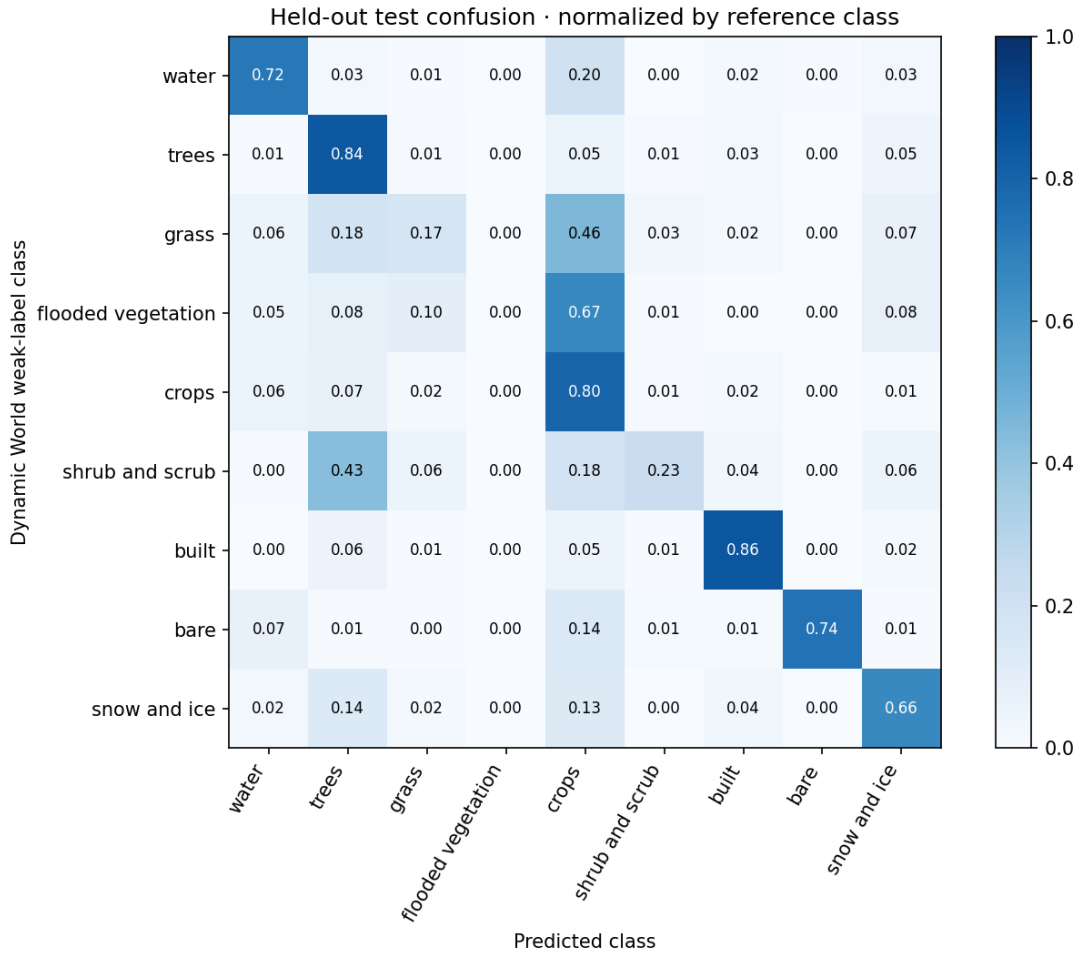


Figure 2. Saved test confusion visualization. Read alongside per-class IoU and support; a dominant class can mask failures in aggregate accuracy.

Held-out region	OA	mIoU
dalarna se	78.34%	38.19%
east oweinat eg	92.21%	21.21%
leipzig de	74.11%	35.66%
ludhiana in	80.94%	23.22%
nairobi ke	54.74%	23.15%
seville es	70.91%	31.28%
sorriso br	50.34%	21.83%

Global mIoU is computed from pooled confusion counts, not by averaging regional mIoUs. Regional support and absent classes differ, so regional means need not reproduce the global value. Pixel-months are correlated observations; the large pixel count must not be treated as millions of independent experimental replicates.

05 Prototype and area accounting

Stage	Implementation and output
Request	Latitude, longitude and first historical month; fixed UTM tile
Retrieval	Twelve monthly Sentinel-1 composites; cache and acquisition receipts
Preprocessing	Frozen terrain/filter recipe, masks and training normalization
Inference	Fixed FP32 checkpoint; nine-class predictions and entropy
Exports	12 maps, georeferenced rasters, shares, transitions and case report
Interaction	Jupyter input form and month slider; optional Streamlit interface

No optical image enters model inference. WorldCover is retrieved separately for an annual external comparison. Entropy is exported as uncalibrated model ambiguity, not as a measured probability that a prediction is wrong.

Missing data is explicit

Monthly percentages use the entire fixed AOI, with an unobserved category. Observed-only percentages are also exported. Class area equals its pixel count multiplied by 100 m^2; these are projected-grid areas rather than geodesic surface areas.

Changes use common support

For a pair of months, comparison is restricted to pixels observed at both dates. For class c, the net change in percentage points is 100 × (later class count - earlier class count) / common pixel count. If no common pixels exist, the change is undefined rather than zero. Nine-by-nine transition matrices retain opposing transitions that a net change can hide.

Mannheim case configuration

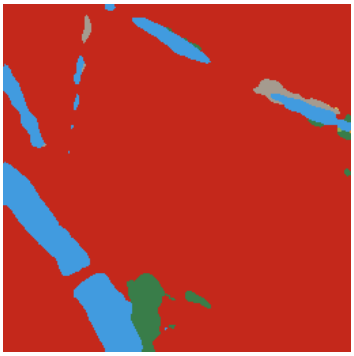
Parameter	Value
Location	49.4875° N, 8.4660° E
Window	January-December 2021
Grid	EPSG:32632; 256 × 256 at 10 m
Projected AOI	6.5536 km^2
Monthly observed coverage	100% in each of the 12 months
Observed pixel-months	786,432 of 786,432
Execution	Inference and external comparison completed

Georeferenced rasters and the full case export are generated on the server. The compact review archive used for this report contains the maps, metrics and tables, but excludes raw arrays, model weights and GeoTIFFs.

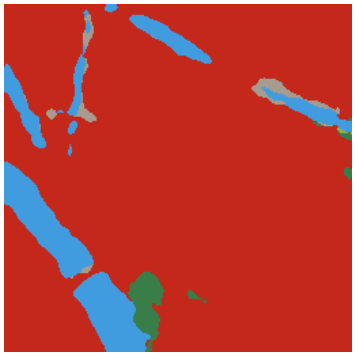
06 Mannheim monthly map atlas

January-December 2021. All maps use the same 2.56 km-wide grid and class palette. Differences are model predictions, not verified land-use changes.

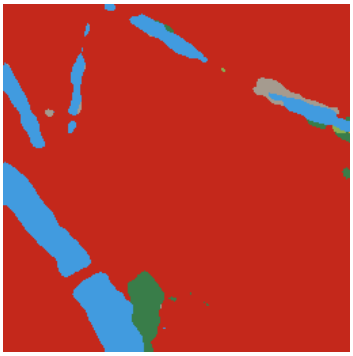
2021-01



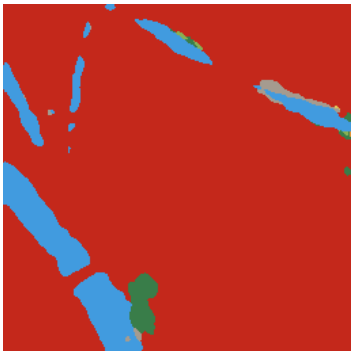
2021-02



2021-03



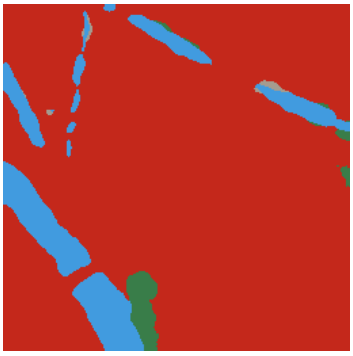
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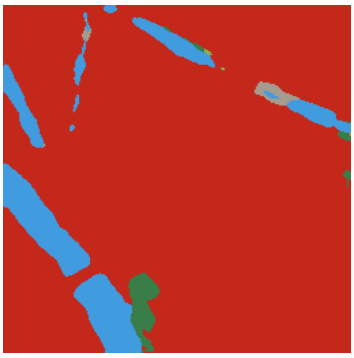
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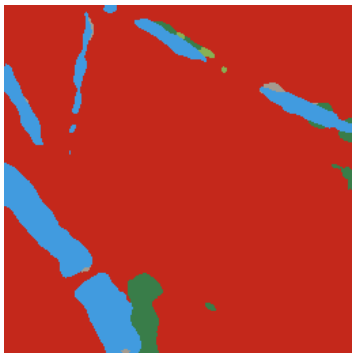
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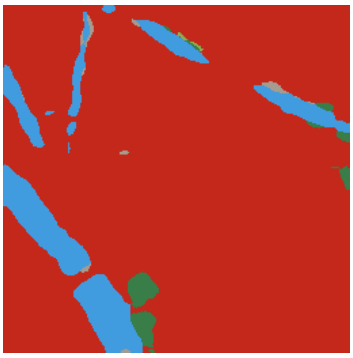
2021-07



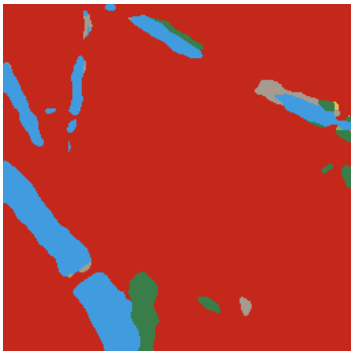
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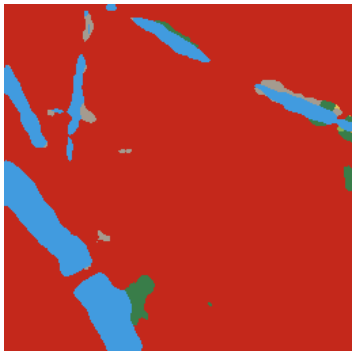
2021-09



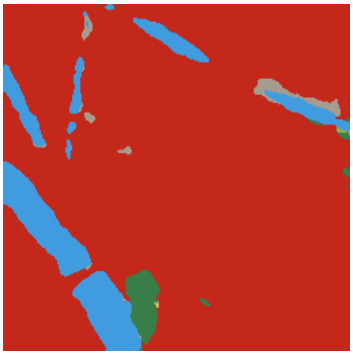
2021-10



2021-11



2021-12



Palette: blue = water; dark green = trees; light green = grass; purple = flooded vegetation; orange = crops; tan = shrub/scrub; red = built; gray = bare; violet = snow/ice.

07 Composition and predicted change

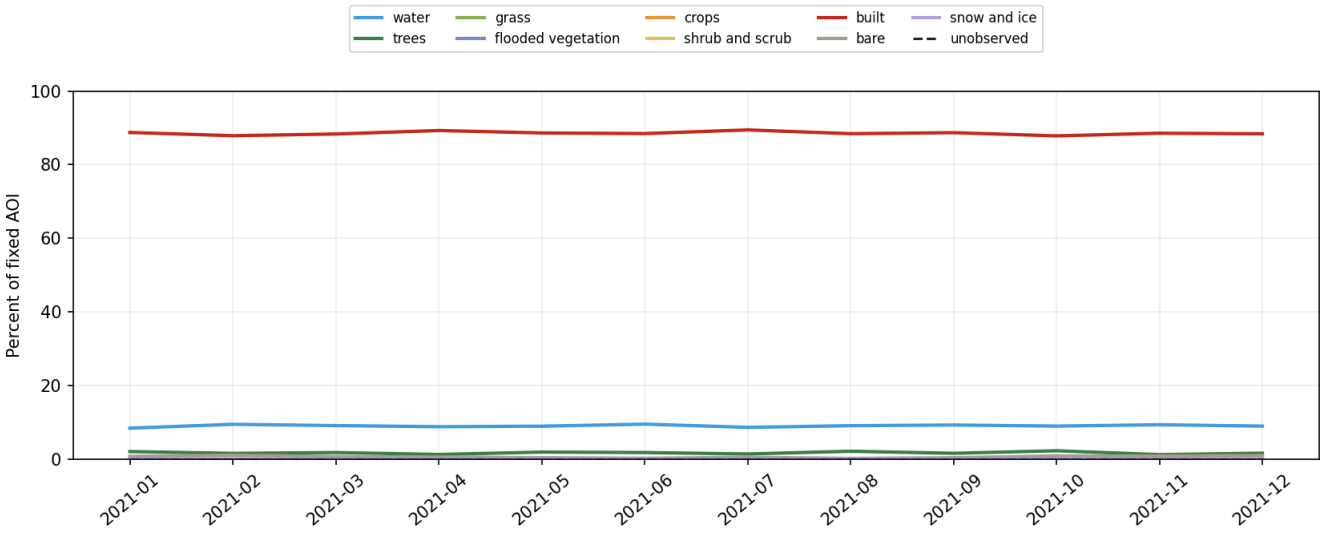


Figure 3. Predicted monthly composition. Built-up land dominates every month. Full coverage means the observed-only and full-AOI percentages coincide for this case.

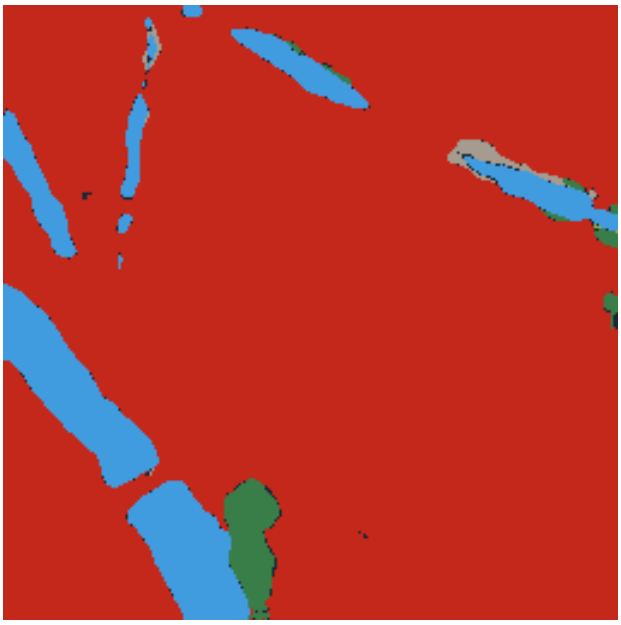
Class	January share	December share	Net change (pp)
water	8.412%	8.971%	+0.558
trees	2.058%	1.619%	-0.439
grass	0.012%	0.031%	+0.018
flooded vegetation	0.000%	0.000%	+0.000
crops	0.003%	0.006%	+0.003
shrub and scrub	0.006%	0.011%	+0.005
built	88.759%	88.390%	-0.369
bare	0.749%	0.974%	+0.224
snow and ice	0.000%	0.000%	+0.000

January-to-December changed-pixel fraction: 1.857% on 100% common support. Water increases by approximately 0.558 percentage points and built decreases by 0.369 percentage points in the predictions. These figures describe classification differences, not demonstrated flooding, construction, deforestation or permanent land conversion.

There is no independent monthly reference in this submission. Acquisition geometry, seasonal conditions, wet surfaces and errors at either date can produce apparent change. Small net changes can also coexist with larger opposing transitions.

08 External comparison and limitations

Annual modal prediction



ESA WorldCover 2021 crosswalk

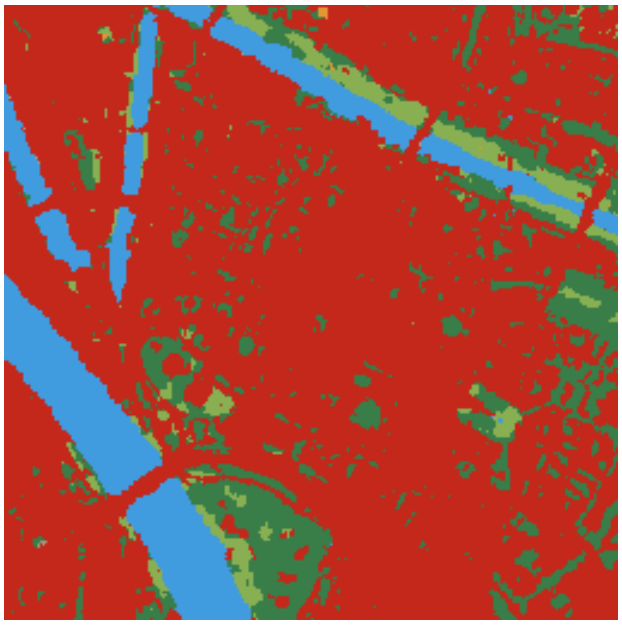


Figure 4. Same-grid annual comparison. Red is built-up land and green is vegetation. The predicted map visibly omits much of the vegetation represented in the external map.

Common annual support	Prediction	WorldCover
Built-up share	88.91%	72.43%
Tree share	1.62%	13.56%

Annual external-map agreement is 80.94% OA but only 26.30% mIoU. Built-up land is overpredicted by about 16.49 percentage points on comparable support. This case demonstrates why overall accuracy alone is insufficient for an urban AOI.

The annual prediction uses the unique modal class across all twelve valid months. Comparison includes 65,303 of 65,536 pixels (99.64%); temporal ties are excluded. WorldCover v200 is a 2021 annual product based on Sentinel-1 and Sentinel-2 [3]. Its broad class crosswalk excludes mangroves and moss/lichen; herbaceous wetland is not identical to flooded vegetation. Neither map is ground truth, and shared sensor families limit independence. An annual comparison cannot validate month-specific change.

Readiness assessment

The system is technically capable of inference and suitable as an assessed research prototype. Its weak classes and local vegetation errors preclude claims of reliable operational land-cover monitoring. Further work should prioritize label quality, class-specific diagnostics, orbit handling and calibration, followed by replicated validation before any operational use.

09 Reproducibility and references

The executed results notebook completed four code cells without recorded errors. The validation replay matched the retained confusion matrix exactly. Saved evaluation and case completion records bind the reported results to the same checkpoint. The interactive notebook is supplied separately for a live input demonstration; saved results do not by themselves verify its browser controls.

Evidence	Package location
Executed results notebook	results/notebooks/02_Results.ipynb
Interactive prototype notebook	results/notebooks/01_Run_Prototype.ipynb
Metrics, protocol and exact replay	results/evaluation/
Offline maps, tables and case report	results/cases/4c50a61fa03e274d/
Source and numerical checks	SARScope_Submission.zip
Environment and installed versions	results/environment.json; requirements.lock.txt

Artifact identity

Checkpoint SHA-256:
a52e9f4f69d4647dfad7b6b01fb3613070f91201aa4dceb004db410f6b79b2b5

Dataset identity:
6c3b62d7d71f22d4d098b9fbaf497daf45fe5c3c4e2198fa62cb1db17c4662cf

Weights, prepared dataset arrays and server GeoTIFF exports are separate large artifacts. Their submission or assessor-accessible links must accompany this compact report package. Do not distribute Earth Engine credentials. Original research backups remain separate from this delivery.

References

[1] Google Earth Engine. Sentinel-1 Algorithms.
https://developers.google.com/earth-engine/guides/sentinel1

[2] Google Earth Engine Data Catalog. Dynamic World V1.
https://developers.google.com/earth-engine/datasets/catalog/GOOGLE_DYNAMICWORLD_V1

[3] Zanaga et al. (2022). ESA WorldCover 10 m 2021 v200.
https://doi.org/10.5281/zenodo.7254221

[4] Ronneberger, Fischer and Brox (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation.
https://arxiv.org/abs/1505.04597

[5] He, Zhang, Ren and Sun (2015). Deep Residual Learning for Image Recognition.
https://arxiv.org/abs/1512.03385

Source documentation checked 9 November 2026. Quantitative findings are calculated from the supplied run artifacts; no benchmark superiority claim is made.